



Muhammad Asad

SOFTWARE ENGINEERING STUDENT

CONTACT

Karachi, Pakistan
0313-2164250
asad.spartan300@gmail.com
linkedin.com/in/muhammadasad-arshad
github.com/asad594

SKILLS

C# & .NET Framework



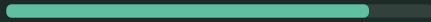
Java



Python



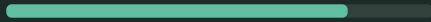
Django & DRF



FastAPI



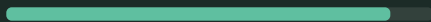
HTML / CSS / JS



SQL & PostgreSQL



OOP & DSA



Git & GitHub



TOOLS & IDES

Visual Studio

IntelliJ IDEA

VS Code

PostgreSQL / MySQL Workbench

Docker (Basics)

Git / GitHub

LANGUAGES

English

Professional

Urdu

Native

Muhammad Asad

Software Engineering Student | Bahria University, Karachi | Batch 2028

PROFILE

Software Engineering undergraduate (Batch 2028) with hands-on experience building REST APIs using Django REST Framework, FastAPI, PostgreSQL, and JWT-based authentication, alongside a strong foundation in desktop application development using C# (.NET Framework), Java, and Python. Currently a Backend Development Intern at ByteCorp Technologies, building a production-style job board backend. Passionate about Object-Oriented Programming, Data Structures, and Algorithm Analysis, with a track record of independently shipping 15+ full-stack, AI-powered, and algorithmic projects.

EXPERIENCE

Backend Development Intern (ByteCorp Technologies)

Jul 2026 - Present

Hybrid - Training Program

- Building a production-style Django REST Framework job board backend: PostgreSQL schema, JWT authentication (15-min access / 7-day refresh), and role-based access control across 3 roles.
- Delivered Phase 1 on schedule; currently implementing strict validation across all modules following a structured PR-based workflow with mentor review.

Java Development Intern (AICTE OIBSIP)

2026

Remote - Java Development Track

- Completing a 5-task Java internship track; built an ATM Interface (Java OOP, ArrayList) and an Online Examination System (Java Swing, timer-based flow).

Software Engineering Intern (CodeAlpha)

2024

Remote

- Developed a Python & Streamlit AI-powered CookBook web application with intelligent recipe discovery and ingredient-based filtering.
- Built task automation scripts in Python that reduced repetitive manual workflows significantly.

PROJECTS

ResumeAI

2026

FastAPI, Next.js, PostgreSQL, JWT, OpenAI/OpenRouter API, Docker

- Built a production-ready AI resume analysis platform with ATS scoring, skill-gap analysis, and job matching; 18+ REST API endpoints and Docker Compose deployment.

FoodHeaven - Restaurant POS System

2026

Python, Django REST Framework, PostgreSQL, PyWebView

- Built a native desktop POS system with real-time cart management, GST calculation, Kitchen Monitor Display, and a Chart.js analytics dashboard.

HR Management System

2025

C# . .NET Framework . Windows Forms

- Engineered a full-featured desktop application to streamline HR workflows including recruitment pipelines and reporting dashboards.
- Implemented a weighted job-ranking algorithm to filter and score candidates automatically by Experience, Skills, and Education.

ADDITIONAL SKILLS

React (Learning)
Docker (Basics)
MIPS Assembly
Digital Logic Design
System Design
Requirement Elicitation
Role-Based Access Control (RBAC)

LINKS

github.com/asad594
linkedin.com/in/muhammadasad-arshad
Portfolio

Karachi, Pakistan

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Projects (continued)

PROJECTS (continued)

KernelFlow - CPU Scheduling Simulator 2026

React (Vite), Tailwind CSS, FastAPI, Python, Docker

- Built a full-stack OS CPU scheduling simulator implementing 6 algorithms (FCFS, SJF, SRTF, Round Robin, Priority) with a React/Vite glassmorphism UI and FastAPI backend.
- Designed an interactive tick-by-tick Gantt chart, live ready-queue visualizer, and analytics dashboard; containerized with Docker Compose.

Restaurant Management System 2025

Java . Swing . OOP Design Patterns

- Built a comprehensive restaurant management solution covering table reservation, order processing, inventory tracking, and billing.
- Applied OOP design patterns (Factory, Observer) to ensure modularity, maintainability, and scalability across all modules.

AlgoPuzzleBoard 2025

C# . Algorithm Visualization . Windows Forms

- Created an interactive educational platform visualizing N-Queens, Knight's Tour, and backtracking algorithms step by step.

CookBook Web App (CodeAlpha Internship) 2025

Python . Streamlit . AI

- Delivered an AI-assisted recipe discovery platform with smart ingredient-based filtering and a clean, responsive UI.

Requirement Genie 2025

Python . NLP . AI

- Built an AI-driven tool that automates software requirement gathering and generates structured SRS documents using NLP techniques.

Decode & Discover 2025

Python . Cryptography . Educational Tool

- Developed an interactive cryptography learning tool with step-by-step visualizations of classic ciphers (Caesar, Vigenere, substitution).

CERTIFICATIONS & ACHIEVEMENTS

- Certified Software Engineering Intern - CodeAlpha (2024)
- Currently completing a Backend Development Internship at ByteCorp Technologies and a Java Development Internship under AICTE OIBSIP (both ongoing).
- Participant in internal coding hackathons and logic-building competitions - Bahria University
- Active open-source contributor: github.com/asad594 (15+ projects)
- Additional projects (Hotel Reservation System, Student Grade Tracker, MIPS Productivity Dashboard, Snake Game, and 10+ mini-projects) available on GitHub.